LAB PRACTICAL # 03

Linux Directory and File Commands and I/O Redirection

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Program	BS-SE (5th Semester)

1. Objective

The objectives of this lab are:

- To learn basic Linux commands for navigating the file system.
- To display the contents of directories.
- To understand absolute and relative paths.
- To learn commands for creating, copying, moving, renaming, and deleting files and directories.
- To understand file handling and I/O redirection in Linux.

2. Software Required

- Linux Operating System
- Linux Terminal

3. Lab Activity 1: Linux Navigation Commands

The following tasks were performed in the Linux terminal to navigate the file system and list directory contents.

Task	Description	Command	Purpose
1	Display the current directory	`pwd`	Displays the current working directory.
2	Change to the /etc directory	`cd /etc`	Changes the current working directory to /etc.
3	Go to the parent directory	`cd ..`	Moves one level up to the parent directory.
4	Go to the root directory	`cd /`	Changes the current working directory to the root directory.
5	List the contents of the root directory	`ls /`	Displays the files and directories present in the root directory.

Task	Description	Command	Purpose
6	Long listing of the root directory	`ls -l /`	Displays detailed information about files and directories in the root directory.
7	List /etc without changing directory	`ls /etc`	Displays the contents of /etc while keeping the current working directory unchanged.
8	List the contents of /bin and /sbin	`ls /bin /sbin`	Displays the contents of both directories.
9	List the home directory	`ls ~`	Displays the files and directories in the user's home directory.
10	List all files, including hidden, in home	`ls -a ~`	Displays all files and directories, including hidden files.
11	List /boot in human-readable format	`ls -lh /boot`	Displays detailed information about files in /boot with sizes in a human-readable format.

4. Important Linux Commands

The table below summarizes the key commands used for file and directory management.

Command	Purpose
`pwd`	Displays the current working directory
`ls`	Lists files and directories
`cd`	Changes the current directory
`mkdir`	Creates a directory
`cp`	Copies files and directories
`mv`	Moves or renames files and directories
`rm`	Removes files and directories
`rmdir`	Removes empty directories
`touch`	Creates an empty file
`cat`	Displays or combines file contents
`less`	Views file contents page by page
`file`	Identifies the type of a file

5. I/O Redirection

I/O redirection is used to change the input or output destination of a command. The main operators are summarized below.

Operator	Example	Description
Output redirection (>)	`ls > output.txt`	Saves the output of ls into output.txt, replacing its previous contents if the file already exists.
Append output (>>)	`echo "New line" >> output.txt`	Appends new text to the end of output.txt.
Input redirection (<)	`cat < output.txt`	Reads the contents of output.txt through standard input and displays them.
Pipeline (|)	`ls /etc | less`	Passes the output of ls /etc to less for page-by-page viewing.

6. Result

The Linux navigation commands were studied and used to explore the file system, display directory contents, and understand the basic use of Linux file and directory commands.

The concepts of file handling and I/O redirection were also studied.

7. Conclusion

In this lab, I learned the basic Linux commands used for navigating the file system and listing directory contents. I also studied file and directory management, along with the concepts of standard input, standard output, and pipelines in Linux.